

Table S1b: Model GWL Crossing Years and Storyline Assignments (SSP2-4.5)

Model	Scenario	GWL +2C Year	Winter SL (+2C)	Summer SL (+2C)
ACCESS-CM2	SSP2-4.5	2041	Decreasing Frequency	Increasing Frequency
ACCESS-ESM1-5	SSP2-4.5	2046	Increasing Frequency	Increasing Frequency
CESM2	SSP2-4.5	2044	Increasing Frequency	Increasing Frequency
CESM2-WACCM	SSP2-4.5	2040	Increasing Frequency	Decreasing Frequency
CMCC-CM2-SR5	SSP2-4.5	2039	Neutral / Other	Increasing Frequency
CMCC-ESM2	SSP2-4.5	2041	Increasing Frequency	Decreasing Frequency
CanESM5	SSP2-4.5	2025	Decreasing Frequency	Decreasing Frequency
EC-Earth3	SSP2-4.5	2045	Neutral / Other	Decreasing Frequency
EC-Earth3-CC	SSP2-4.5	2036	Decreasing Frequency	Neutral / Other
EC-Earth3-Veg-LR	SSP2-4.5	2050	Decreasing Frequency	Decreasing Frequency
GFDL-ESM4	SSP2-4.5	2074	Decreasing Frequency	Decreasing Frequency
HadGEM3-GC31-LL	SSP2-4.5	2034	Increasing Frequency	Increasing Frequency
IITM-ESM	SSP2-4.5	2062	Decreasing Frequency	Decreasing Frequency
INM-CM4-8	SSP2-4.5	2064	Increasing Frequency	Increasing Frequency
INM-CM5-0	SSP2-4.5	2073	Neutral / Other	Neutral / Other
IPSL-CM6A-LR	SSP2-4.5	2034	Neutral / Other	Neutral / Other
KACE-1-0-G	SSP2-4.5	2024	Increasing Frequency	Increasing Frequency
KIOST-ESM	SSP2-4.5	2041	Neutral / Other	Increasing Frequency
MIROC-ES2L	SSP2-4.5	2064	Decreasing Frequency	Decreasing Frequency
MIROC6	SSP2-4.5	2074	Increasing Frequency	Neutral / Other
MPI-ESM1-2-HR	SSP2-4.5	2064	Increasing Frequency	Increasing Frequency
MPI-ESM1-2-LR	SSP2-4.5	2058	Decreasing Frequency	Decreasing Frequency
MRI-ESM2-0	SSP2-4.5	2050	Decreasing Frequency	Decreasing Frequency
NorESM2-MM	SSP2-4.5	2079	Increasing Frequency	Increasing Frequency
UKESM1-0-LL	SSP2-4.5	2034	Decreasing Frequency	Neutral / Other